CS 498 QC ✧ Spring 2026

Homework 2

Due Tuesday, March 10th, 2026 at 9pm

- For this homework, you are encouraged to collaborate with your classmates. You should submit your solutions **individually**.
- **Submit your solutions electronically on Gradescope as PDF files.**
- Please format your solutions so that each problem begins on a new page and your name appears at the top of each page. While handwritten submissions are allowed, all homework submissions are **preferred to be typeset** using Latex or other equivalent programs.
- Problems marked ★ are optional and usually a little harder.

☞ Some important course policies ☜

- You are encouraged to collaborate with your classmates on homework problems. You are allowed to look at definitions of mathematical objects and general problem solving strategies as well. You should first make a serious attempt to solve the problems on your own. After that, you may use AI tools like ChatGPT, and you are encouraged to experiment with them.

Note: The goal of these homework problems is to help you better understand the strangeness of entanglement, as well as various applications of it that we saw in the lectures. In particular, we will revisit the EPR paradox, consider a non-local game similar to the CHSH game and revisit quantum teleportation.

Problems:

1. (10 points) **EPR paradox.**

   Let us examine the properties of the EPR pair

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle .$$

   (a) Let $A = \{|a_0\rangle, |a_1\rangle\}$ be some orthonormal basis for $\mathbb{C}^2$. Suppose Alice measures her qubit using basis A. What are the probabilities to obtain each measurement outcome?

   (b) Show that if Alice obtains measurement outcome $|a_i\rangle$ for $i \in \{0,1\}$, the post-measurement state of the EPR pair is $|a_i\rangle \otimes |a_i\rangle^*$ where $|a_i\rangle^*$ is the complex conjugate of $|a_i\rangle$ (i.e. the j-th entry is the complex conjugate of the j-th entry of $|a_i\rangle$).

   This is interesting because Alice might have decided on the basis only after Bob was sent away, yet Alice's measurement causes Bob's qubit to instantaneously collapse into one of the basis states of A (up to complex conjugation). This is a phenomenon called *quantum steering*, because Alice is able to steer Bob's qubit, even though she is only acting on her qubit.

   (c) Suppose that Bob then measures his qubit using an orthonormal basis $B = \{|b_0\rangle, |b_1\rangle\}$. What is the probability distribution of his measurement outcomes, conditioned on Alice's outcome?

   (d) Suppose the order of measurements were reversed: Bob measures his qubit first using basis B, and then Alice measures her qubit using basis A. Show that the joint probability distribution of their outcomes is the same as before.

   (e) What can you conclude about the effectiveness of using quantum entanglement and quantum steering as a method for faster-than-light communication? In other words, can Alice and Bob, by only making local measurements on their entangled state, send information to each other?

2. (10 points) **A Magic Non-local Game.** Alice, Bob and Charlie play the following game. A referee samples a question uniformly at random from the set $\{000, 110, 101, 011\}$, and sends the first bit of the question to Alice, the second to Bob, and the third to Charlie. Each of them is required to respond with a single bit $\{0,1\}$. Denote the bits of the question as x, y, z, and the answers as a, b, c for Alice, Bob and Charlie respectively. The three players win if

$$a \oplus b \oplus c = x \vee y \vee z .$$

   The three players can meet to decide on a strategy *before* the game begins, but are *not allowed to communicate* once the questions are sent.

(a) Show that the optimal classical winning probability (deterministic or randomized) is $\frac{3}{4}$.

(b) Consider the following quantum strategy: Alice, Bob and Charlie share (before the game begins) the three-qubit state

$$\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|111\rangle ,$$

where each player keeps one of the three qubits. Each player then acts as follows:

- Upon receiving question 0, the player measures their qubit in the basis

$$\{|+\rangle , |-\rangle\} ,$$

and responds with the measurement outcome.

- Upon receiving question 1, the player measures their qubit in the basis

$$\{\frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle , \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle\} ,$$

and responds with the measurement outcome.

Compute the success probability of this strategy and show that this is the optimal quantum strategy for this game.

3. (10 points) **Quantum Teleportation with Noise.** We saw how to teleport quantum states in class. Let's consider a twist on the standard teleportation protocol. Let's imagine that when Alice and Bob meet up to create an entangled state, the settings on their lab equipment were screwed up and they accidentally create the following two-qubit entangled state

$$|\theta\rangle = \frac{1}{\sqrt{3}}|00\rangle - \frac{1}{\sqrt{6}}|01\rangle + \frac{1}{\sqrt{6}}|10\rangle + \frac{1}{\sqrt{3}}|11\rangle .$$

Alice takes the first qubit and Bob the second qubit of the above state $|\theta\rangle$, but only Alice realizes that there was a screw up in the lab (for instance by seeing that the equipment was not calibrated correctly) and this was after Bob has gone his separate way. Suppose that Alice now gets a gift qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and she still wants to teleport $|\psi\rangle$ to Bob, using their corrupted entangled state $|\theta\rangle$ and a classical communication channel they share.

Show how the teleportation protocol can be adapted for the corruption from Alice's side and analyze the correctness of your proposed protocol. Like in the standard teleportation protocol, Alice can only apply unitaries and measurements to her two qubits, and Bob will behave exactly the same as in the standard teleportation protocol (since he's not aware of the corruption).

4. (20 points) **Transferring Entanglement.** Let's say there are three parties, Alice, Bob, and Carol. Alice shares an EPR pair with Bob, and Bob shares an EPR pair with Carol (so Alice has one qubit, Bob has two qubits, and Carol has one qubit). Design and analyze a protocol that involves only classical communication between the pairs (Alice, Bob), and (Bob, Carol), such that at the end Alice and Carol — who never directly interacted with each other — now share an EPR pair. *Hint: use the teleportation protocol as inspiration.*